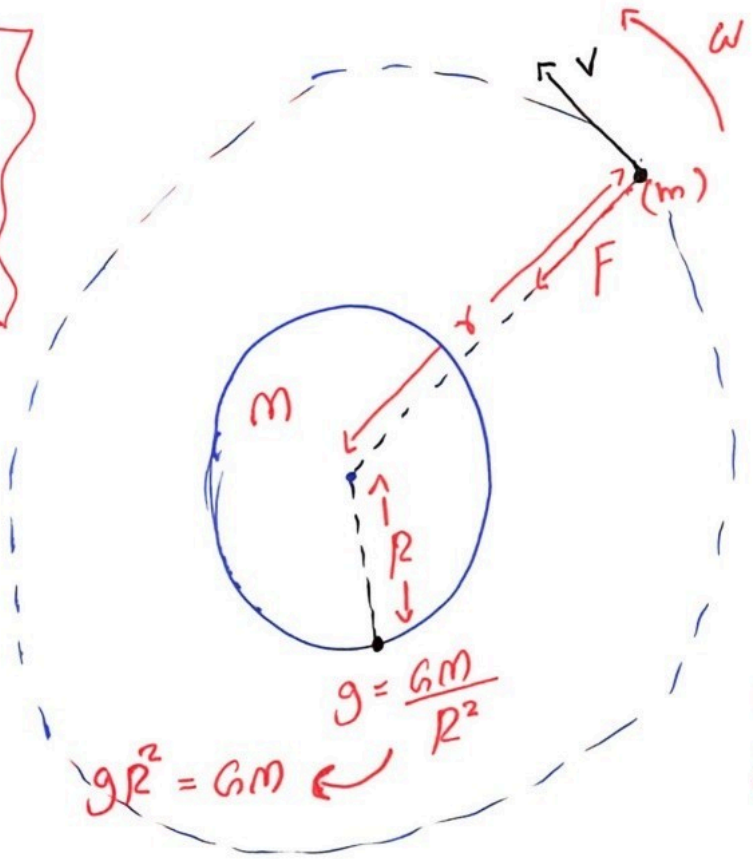
Gravitational Field

⇒ Circular motion of an object around a planet

$$a_c = \frac{v^2}{r}$$

$$a_c = \omega^2 r$$



$$g = \frac{GM}{R^2}$$

$$gR^2 = GM$$

$$v = \sqrt{\frac{gR^2}{r}}$$

$$F_c = m a_c$$

$$\Downarrow$$

$$F = m\frac{v^2}{r}$$

$$\frac{GMm}{r^2} = m\frac{v^2}{r}$$

$$\Downarrow$$

$$v = \sqrt{\frac{GM}{r}}$$

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$$\frac{GMm}{r^2} = mr\omega^2$$

$$\Downarrow$$

$$\frac{GM}{r^2} = r\left(\frac{2\pi}{T}\right)^2$$

$$T^2 = \left(\frac{4\pi^2}{GM}\right) r^3$$

$$\Downarrow$$

$$T^2 \propto r^3$$

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_1^3}$$

$$\omega = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{\omega}$$

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